What makes an asset the dominant intermediary?

Centre for Blockchain Technologies

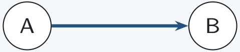
The Making of Dominant Vehicle Currencies

Evidence from DeFi

Jiahua Xu

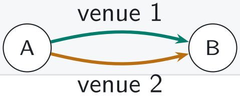
How we reconstruct a route from one transaction

Direct swap
one transaction

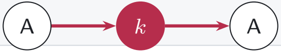


Sequential vehicle route
one transaction

Parallel split
one transaction



Round trip
one transaction

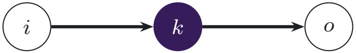


Vehicle choice directs indirect order flow toward particular assets and pools. Transaction-level paths reveal that choice rather than leaving it inside aggregate turnover.

Vehicle dominance is the share of indirect trade carried by an asset

MEASUREMENT

- Dominance is the share of indirect trade intermediated by asset k.
- Count share treats each route equally; value share weights routes by dollar value.
- Network measures capture how broadly k connects endpoint pairs.

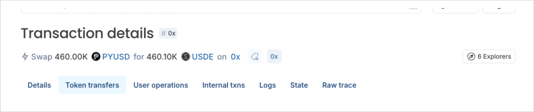


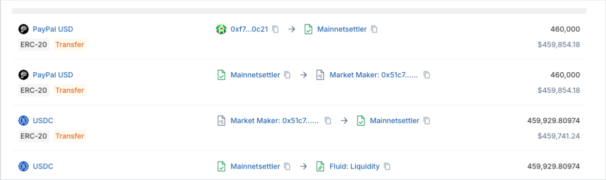
Count-weighted dominance
Share of indirect routes that pass through k

Value-weighted dominance
Share of comparable routed value that passes through k

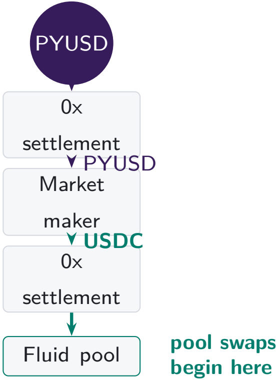
The transaction starts with PYUSD

EXPLORER RECORD





SETTLEMENT HANDOFF



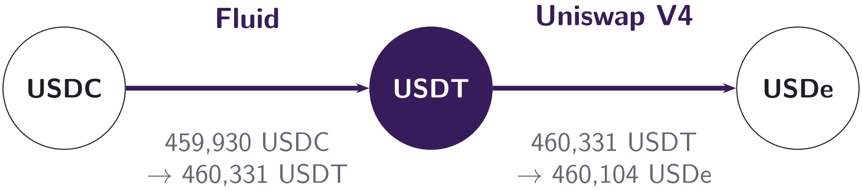
PYUSD reaches 0x settlement; the market maker supplies USDC for execution in the pools.

Note: The explorer screenshot and transfer trace refer to transaction `0xbda1d07f...9bf67ad9` on 10 January 2026.

USDT links the two pool swaps

ORDERED TRANSFERS AND ROUTE RECONSTRUCTION

PYUSD settlement ⟶ **USDC** ⟶ **USDT** ⟶ **USDe**



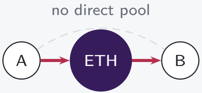
vehicle

The connected component converts USDC into USDe across two pools.
USDT is the vehicle because it links the two legs.

Note: The route is the connected pool-swap component of transaction `0xbda1d07f...9bf67ad9` on 10 January 2026.

Architecture changes routing opportunities

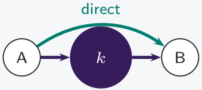
V1 mandatory hub



Only ETH–token pools: token-to-token execution must pass through ETH.

Hub use is imposed

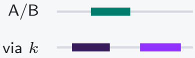
V2 pair choice



Arbitrary pairs add a direct route; vehicle routes remain feasible.

Hub use becomes a choice

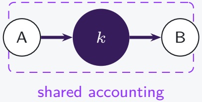
V3 range choice



LPs choose ranges and fees; active depth can favor direct pairs or vehicle spokes.

Opportunity shifts by pair

V4 net settlement



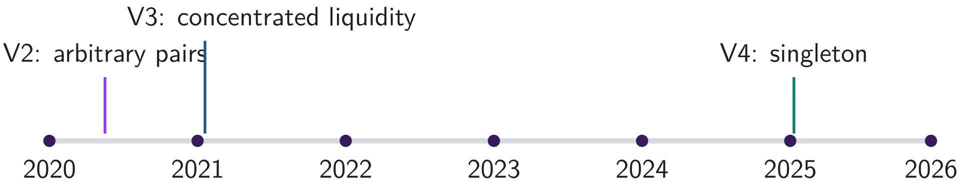
Two pools still execute; intermediate token movement may be netted.

Settlement can diverge from route use

Architecture feasible rules **Exposure** realised pool and route use **Calendar date** timing only

The route sample and architecture supplement answer different questions

DATA



Main route sample

Swap legs from multiple protocols are linked within each transaction. A leg may come from a two-asset pool or from one asset pair inside a multi-asset Curve or Balancer pool.

V1/V2 architecture supplement

Uniswap V1 forced-ETH paths and V2 pair formation are analysed separately. They answer protocol-design questions and do not define the main route perimeter.

Vehicle dominance aggregates realised intermediary choices

One realised route

k occupies the intermediary position

Vehicle dominance

D_k = use through k
$\div$ total intermediary use

Value-weighted routes use comparable dollar amounts

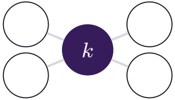
Excess vehicle use

$X_k = D_k \div E_k$

E_k: endpoint share of k

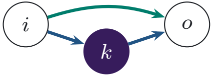
$X_k > 1$: intermediary use exceeds the endpoint benchmark

Separate characteristics of the asset and route set



Network position

position of k in the feasible graph

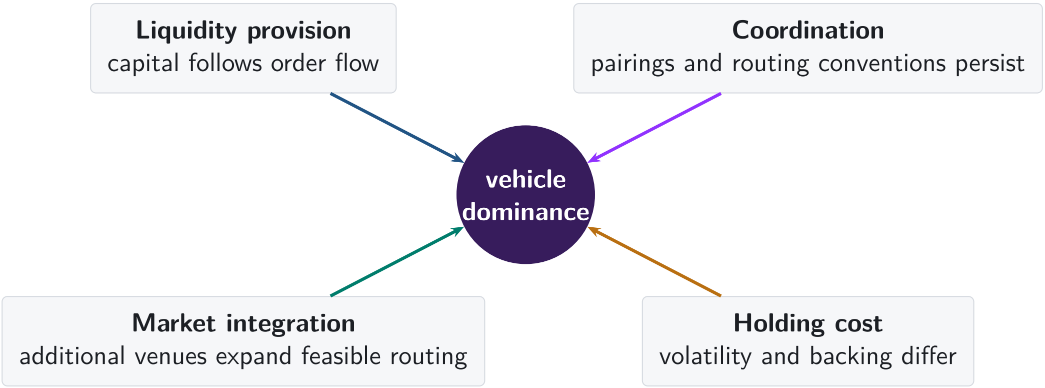


Execution cost

which route returns more at the same state

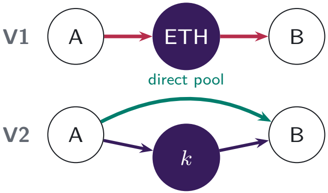
Endpoint demand supplies the benchmark; network position and same-state route costs remain separate characteristics.

Why might one asset become the dominant vehicle?



Protocol design changes markets, depth, and settlement

V1 ⟶ V2: PAIR CREATION



V1 admits only ETH–token pools. V2 permits any ERC-20 pair, so market creation determines whether direct and vehicle paths coexist.

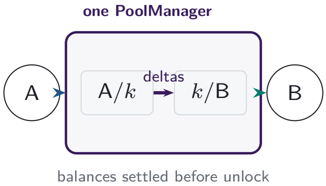
Which edges exist?

V3: ACTIVE LIQUIDITY

LPs choose a price range and fee tier for each pool. Positions covering the current price determine active liquidity in the direct pair and in each vehicle spoke.

Which path offers depth at this size?

V4: SHARED ACCOUNTING

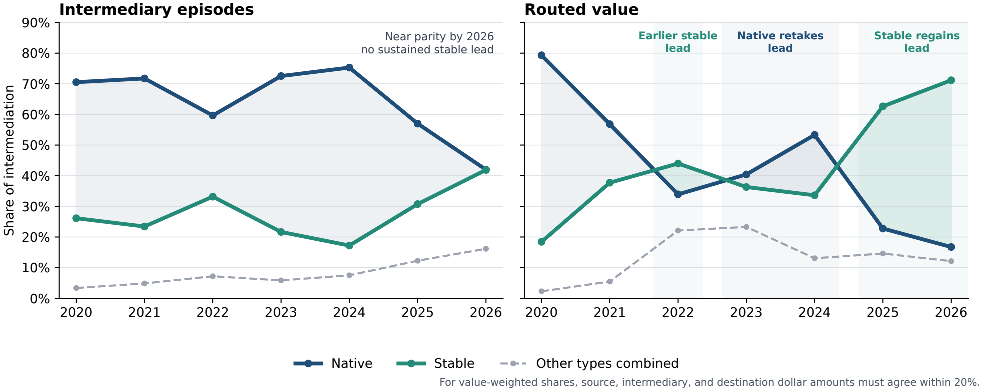


One PoolManager holds all pools. During a lock, the two swaps update balance deltas; the caller settles its obligations before the lock ends.

Where is the route settled?

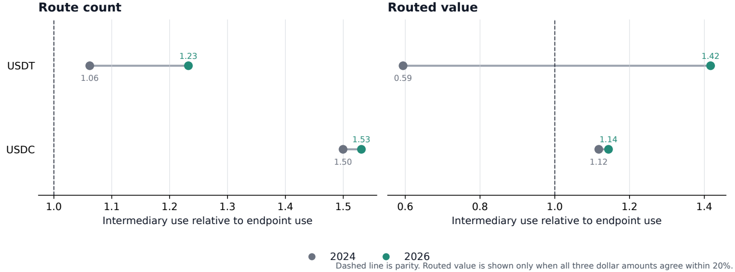
Economic implication. Pair creation changes available paths; concentrated liquidity changes pair-specific execution at the current state and trade size; the singleton changes the settlement boundary for a route that still uses two pools.

Stablecoins regain the lead in routed value



Note: The route universe is exact two-leg routes with an observed intermediary. The left panel counts one intermediary episode per route and divides each asset type's episodes by all intermediary episodes. The right panel divides each type's routed value by total routed value among routes whose source, intermediary, and destination dollar amounts agree within 20%. The 2026 observation covers January–June; Other combines imported, liquid-staking, and residual assets.

USDT drives the rise in stablecoin intermediation



USDT crosses parity by routed value and extends its count advantage; USDC changes little.

Descriptive accounting: 93.7% of the count-share change occurs because USDT's share rises within the single- and cross-venue categories; 6.3% comes from reweighting between them. The corresponding routed-value shares are 85.6% and 14.4%.

Note: Excess use is intermediary share divided by endpoint share on the same route perimeter. The accounting identity uses native assets, USDC, and USDT as its denominator and carries no sampling interval. The comparison uses the same January–June dates in 2024 and 2026.

WETH-linked trading pairs account for most of the count rotation

WETH

WETH is the source or destination

WETH cannot also be the intermediary; stable share is mechanically 100%.

All matched trading-pair groups

$+21.1$ **pp**

95% CI $[+18.2$ pp, $+24.0$ pp$]$

WETH-linked route-count mass: 26.6% $\to$ 46.8%

$\longrightarrow$ exclude WETH endpoints

Trading-pair groups without WETH endpoints

$+3.7$ **pp**

95% CI $[+2.3$ pp, $+5.2$ pp$]$

79,347 matched groups

Within the same trading pair, calendar position, and route scope: all groups $+0.2$ pp $[-1.3$ pp, $+1.7$ pp$]$; without WETH endpoints $+0.3$ pp $[-1.9$ pp, $+2.6$ pp$]$.

The count increase falls from $+21.1$ pp to $+3.7$ pp without WETH endpoints. This is an allocation result:

WETH-linked groups mechanically admit only stablecoin intermediaries in this comparison.

Note: Exact two-leg native-WETH-versus-stablecoin routes on 181 common month-days. Intervals use 30-day Newey–West covariance on 362 endpoint-year daily observations; 2025 is excluded and actual dates preserve the gap. Within-group WLS fixes the trading pair, month-day, and route scope and clusters by pair and date. Feasible routes are not fixed; all comparisons are descriptive.

The routed-value rotation extends beyond WETH-linked trading pairs

Without WETH endpoints

$+21.5$ **pp**

95% CI $[+11.8$ pp, $+31.1$ pp$]$

strict-support stablecoin share of routed value

Full matched sample: $+27.2$ pp $[+23.4$ pp, $+30.9$ pp$]$

$=$

Routed-value reallocation

$+23.5$ **pp**

95% CI $[+15.0$ pp, $+32.1$ pp$]$

Value shifts toward trading-pair and route-scope groups with higher stablecoin shares.

$+$

Change within groups

-2.0 **pp**

95% CI $[-9.8$ pp, $+5.7$ pp$]$

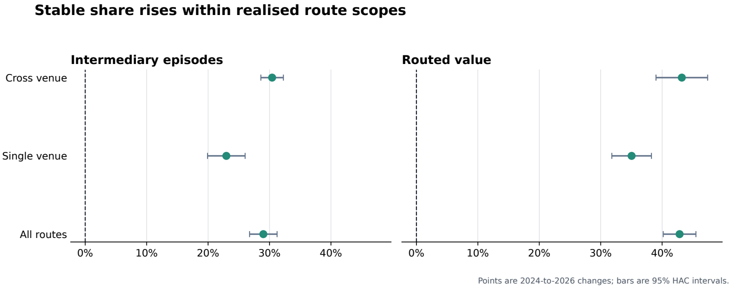
Stablecoin share changes within the same trading-pair and route-scope group.

Exact midpoint identity on each of 181 common calendar days; 2024 versus January–June 2026.

After removing WETH endpoints, value moves toward trading-pair and route-scope groups in which stablecoins already carry more intermediation. The change within those groups is small and imprecise.

Note: Routed value requires source, intermediary, and destination dollar amounts to agree within 20%. The daily midpoint identity is exact. Intervals use the same endpoint-year calendar-HAC covariance as the preceding frame; paired-month-day and bandwidth sensitivities remain in the source. Mukhin (2022), Gopinath and Stein (2021), and Carletti et al. (2021) guide the composition framing, not identification.

Stablecoin use rises within each venue scope

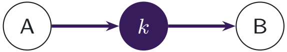


The increase appears within both realised route scopes, so a change in the single- versus cross-venue mix cannot account for the aggregate rotation. Cross-venue status remains selected; the comparison does not identify an effect of integration, routing software, or protocol architecture.

Note: Changes compare the same January–June dates in 2024 and 2026. Bars show 95% calendar-HAC confidence intervals.

Route use, token movement, and LP capital are different objects

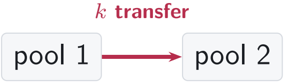
ROUTE INTERMEDIATION



How much indirect trade passes through asset k?

Measure: route or routed-value share.

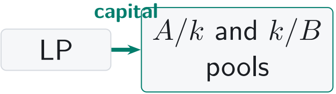
EXTERNAL TOKEN MOVEMENT



How much intermediary token moves between contracts?

Measure: external transfers and transferred value per route.

LIQUIDITY-PROVIDER CAPITAL



How do liquidity stocks, executable depth, and LP reallocations relate to vehicle use?

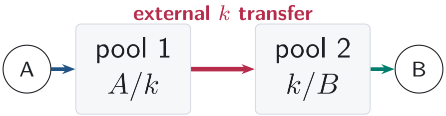
Deposited capital: end-of-day V2 pool reserves in USD.
Executable depth: output at fixed trade sizes and directions.
Provider flows: signed and gross V3 LP dollars over the response window.

Question frame. At the candidate-day level, compare route use with each measure at exact 1-, 7-, 30-, and 120-day horizons.

V2 stocks and V3 flows are estimated separately; depth is not LP capital. Timing alone is not causal.

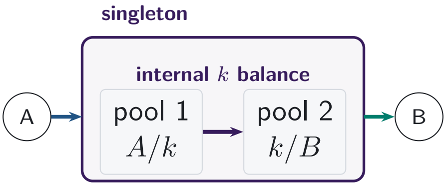
V4 can net settlement without removing the vehicle route

SEPARATE POOL CONTRACTS



The economic route and the intermediary-token movement are both visible in external transfers.

V4 SINGLETON AND FLASH ACCOUNTING



The $A \to k \to B$ route remains observable in pool actions while the intermediary balance is netted internally.

External settlement. Compare like-sized two-pool routes with the same source, intermediary, destination, direction, and week; measure external intermediary-token transfers and value.
Route choice. Compare ordered endpoint-pair–intermediary weeks only when direct and indirect alternatives are both feasible.

LP timing. On pool–intermediary days, relate route use to deposited capital and provider flows at exact 1-, 7-, 30-, and 120-day horizons. Each object is tested separately.

Primary sources: Adams et al. (2021), *Uniswap v3 Core*; Adams et al. (2024), *Uniswap v4 Core*, Sections 1 and 3.

Native-asset pairing persists after it becomes optional

V1: PROTOCOL RULE

217,003 token-to-token trades pass through ETH

Every token pool pairs with ETH, so the architecture imposes the vehicle.

$\Longrightarrow$

V2: MARKET CHOICE

95.5%

of single-leg V2 trades use a WETH pool in 2026

97.9%

of pairs first traded in 2026 include WETH

V2 converted ETH pairing from a protocol rule into a market choice. Native-asset concentration remained high through pair formation and realised trading.

Note: V1 counts linked token-to-ETH and ETH-to-token swaps with matching ETH quantities in one transaction. V2 shares use single-leg Uniswap V2 trades and newly traded pairs; they describe that venue rather than market-wide vehicle dominance.

Stablecoins regained the routed-value lead as trading shifted

WHEN
The value lead changed twice

stablecoin lead	native-asset lead	stablecoin lead
2022	2023–24	2025–26

Stablecoins led routed value, lost the lead, and regained it.

WHICH ASSET
USDT's relative use rose

0.59 ⟶ **1.42**

2024 2026

Excess use $=$ intermediary share $\div$ endpoint share.

HOW ROUTED MARKETS CHANGED
Value shifted across pair groups

$+21.5$ **pp net**

$+23.5$ pp across groups
-2.0 pp within groups

Outside WETH endpoint pairs, stablecoins gained share as value shifted across trading-pair and route-scope groups.

$i \longrightarrow k \longrightarrow o$

The asset in the middle determines which pools carry the indirect trade.

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

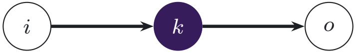
Robustness

- A7 Daily and weekly frequencies
- A8 Count and value weighting
- A9 Endpoint demand

Mechanisms

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

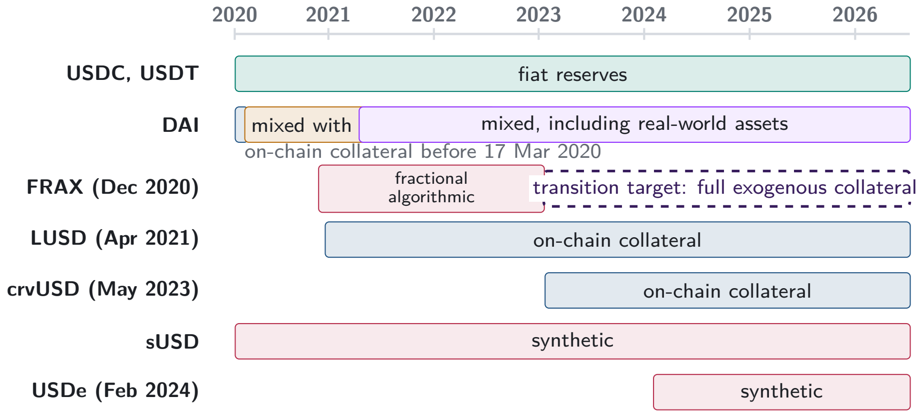
A1. One reconstructed route is one unit



one route unit
two swap legs

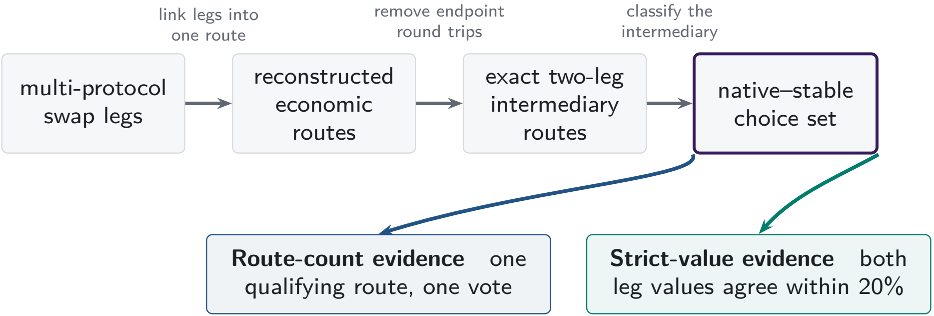
- Each linked $i \to k \to o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k.

A2. Stablecoin backing cannot be treated as fixed



Note: Bars begin at public launch or the sample start for older tokens. DAI's dated regimes follow the Maker Governance actions that added USDC-A and activated RWA001-A. FRAX's dashed arrow follows FIP-188 and denotes a target, not observed collateral.

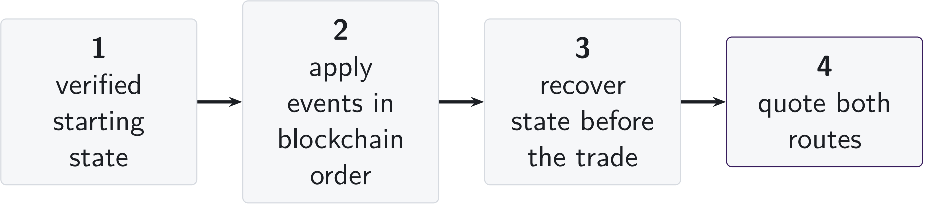
A3. One route universe supports two measurements



Box widths do not represent attrition.

The route sample links legs across protocols; one asset pair inside a multi-asset Curve or Balancer pool may supply a leg. The V1/V2 architecture evidence is separate. Dollar support affects value weighting, not route inclusion.

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hooks
Constant product	■	–	–	–	–	■	–
Concentrated liquidity	■	■	–	–	–	■	–
StableSwap families	■	–	■	–	■	■	–
Weighted pools	■	–	–	■	■	■	–
V4 singleton	■	■	–	–	–	■	□

The common object is the state immediately before execution. The fields required to recover that state vary with the pricing rule.

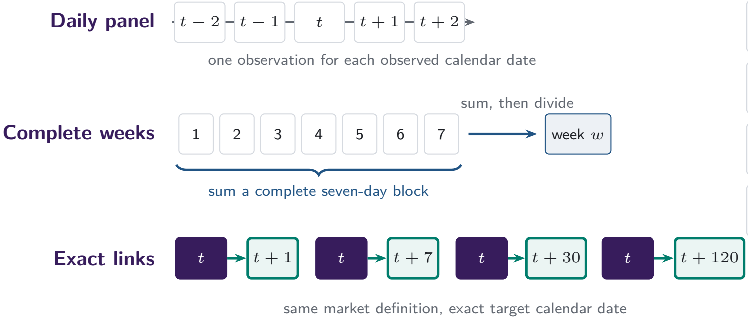
■ required state
□ hooks outside comparison

A6. V1 forced routes have an on-chain signature

- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.

same transaction, matching ETH flow

A7. Daily and weekly frequencies answer different questions



Model role

Daily panel date fixed effects

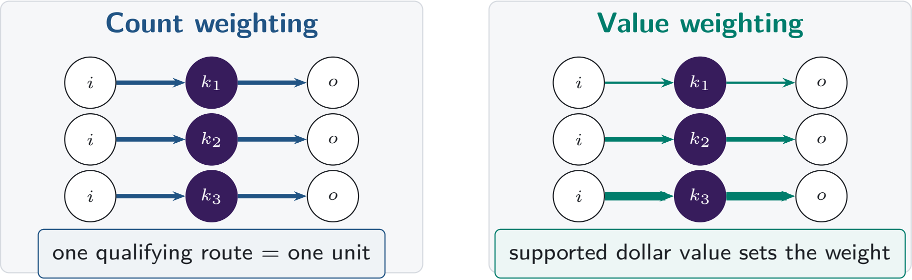
Calendar check week and weekday fixed effects

Complete weeks week fixed effects

Exact links inference by origin date

Missing target dates remain missing: row shifts and calendar months do not substitute for exact calendar links.

A8. Count versus value weighting



Note: Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

A9. Endpoint demand and intermediary use are separate margins

Intermediary share

$$I_{k,t}^{N} = \frac{n_{k,t}^{I}}{\sum_j n_{j,t}^{I}}$$

How often asset k sits inside a route.

Endpoint share

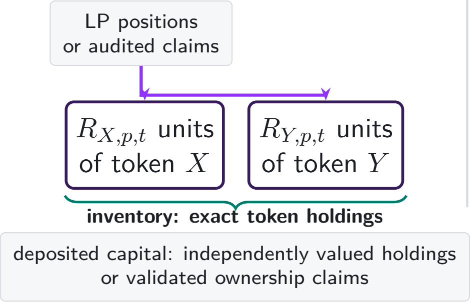
$$E_{k,t}^{N} = \frac{n_{k,t}^{E}}{\sum_j n_{j,t}^{E}}$$

How often asset k is what traders buy or sell.

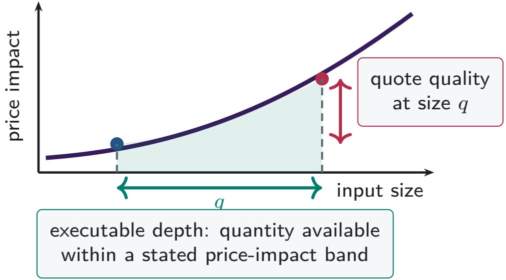
Excess use compares intermediation with endpoint demand on the same support.

A10. Capital, inventory, and depth are different objects

Pool balance sheet at time t

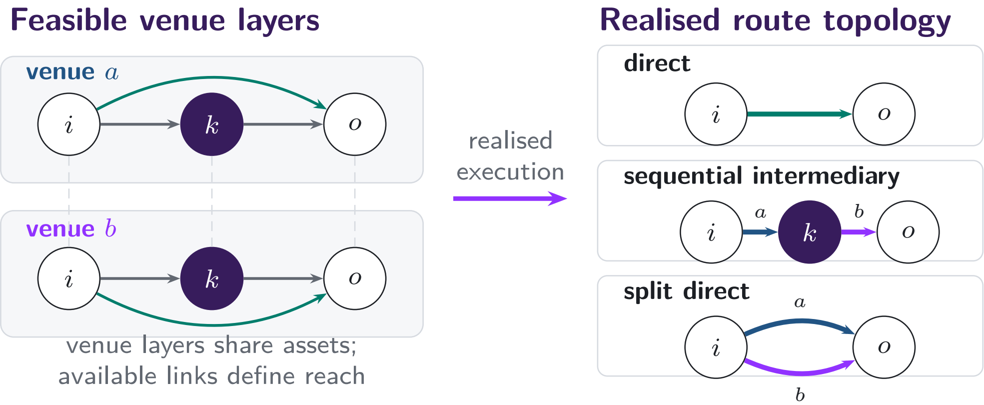


Execution from that state

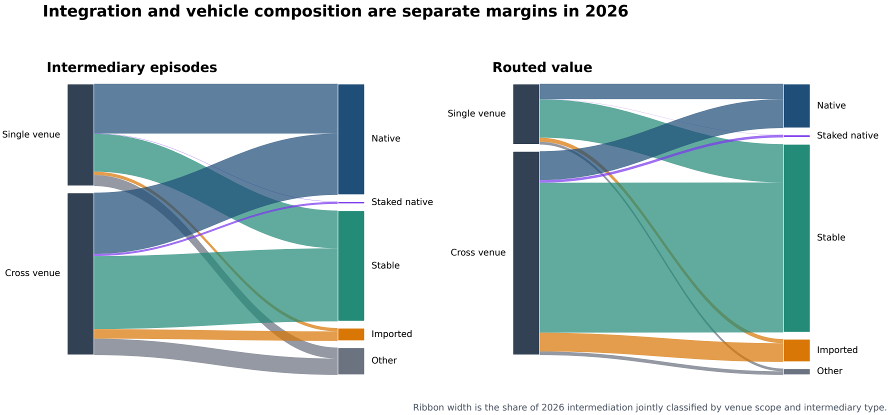


Capital is a stock. Depth and quotes depend on trade size and current pool state.

A11. Additional venues expand the feasible route set



A11b. Venue scope and vehicle type differ in 2026



Note: Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

A12. References: vehicle currencies and market structure

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- Dowd and Greenaway (1993), *Economic Journal*
- Flandreau and Jobst (2009), *Economic Journal*
- Gopinath and Stein (2021), *Quarterly Journal of Economics*
- Krugman (1980), *Journal of Money, Credit and Banking*
- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: decentralised exchange design

- Adams et al. (2021), *Uniswap v3 Core*
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- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
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- Lehar and Parlour (2024), *Journal of Finance*
- Milionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
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